

HOONAH, AK, USA

WMO: 702607

Lat: 58.096N

Lon: 135.409W

Elev: 0

StdP: 14.70

Time Zone: -9.00 (NAK)

Period: 02-19

WBAN: 25378

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.3	13.2	-4.7	4.3	13.1	0.7	5.7	19.6	20.1	31.0	18.2	29.2	6.5	80	0.629

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	74.7	59.6	70.5	58.3	66.3	56.8	61.5	70.9	60.0	67.3	58.6	64.3	5.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.2	72.5	61.5	57.2	69.7	60.9	56.1	67.0	60.1	27.3	70.6	26.3	67.5	25.4	64.2	67.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.0	15.1	12.6	DB	2.6	80.8	4.6	3.9	-0.7	83.7	-3.4	86.0	-6.0	88.2	-9.4	91.0	(4)
(5)			WB	2.4	63.9	4.9	1.8	-1.1	65.2	-3.9	66.3	-6.6	67.3	-10.2	68.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.0	30.5	30.4	32.2	38.8	47.0	53.2	56.9	56.5	50.0	42.0	34.2	31.0	
	DBStd	11.41	7.18	6.98	6.75	4.53	5.14	4.49	3.55	3.53	3.97	4.78	6.61	6.58	
	HDD50	3532	604	548	550	336	121	12	1	1	47	251	473	588	
	HDD65	8407	1069	968	1015	786	557	356	251	264	451	714	923	1053	
	CDD50	602	0	0	0	0	29	107	216	202	46	2	0	0	
	CDD65	3	0	0	0	0	0	1	1	1	0	0	0	0	
	CDH74	158	0	0	0	0	2	63	47	46	0	0	0	0	
	CDH80	22	0	0	0	0	0	10	5	7	0	0	0	0	
Wind	WSAvg	4.5	6.1	4.8	5.5	4.7	4.3	3.8	3.2	3.2	3.4	4.0	5.3	5.6	
Precipitation	PrecAvg	71.4	7.4	5.8	5.2	3.6	3.0	2.4	3.3	4.6	8.5	10.2	8.7	8.7	
	PrecMax	95.4	13.2	9.3	11.7	6.7	6.1	4.4	6.1	8.6	12.0	16.2	16.1	17.1	
	PrecMin	58.0	3.3	2.0	0.8	1.0	0.7	0.9	1.0	1.7	3.0	4.7	2.8	4.5	
	PrecStd	8.9	2.4	2.1	2.2	1.4	1.4	0.8	1.3	1.7	1.9	2.6	2.8	3.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.5	46.2	52.6	63.7	72.5	81.4	80.0	79.3	67.8	55.4	51.5	47.1	
		MCWB	45.7	41.1	44.6	50.3	55.0	61.2	61.4	61.7	55.3	50.4	48.6	43.7	
	2%	DB	43.9	43.5	46.5	56.1	67.6	73.4	73.2	73.3	63.7	53.5	46.5	44.8	
		MCWB	43.1	40.5	41.3	44.7	54.1	58.5	60.7	59.4	55.2	49.8	44.7	42.7	
	5%	DB	41.1	41.7	43.9	52.1	63.3	67.4	68.8	69.3	60.3	51.7	45.0	41.9	
		MCWB	40.3	39.6	39.9	43.8	51.4	56.6	59.4	59.1	53.8	49.1	43.2	40.5	
	10%	DB	39.2	39.5	41.7	48.2	58.9	63.1	64.5	64.4	57.4	49.6	43.1	39.2	
		MCWB	38.5	37.5	38.5	42.4	49.6	54.7	57.5	57.8	53.1	47.2	41.4	38.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.4	43.5	46.1	50.8	58.3	62.2	63.5	63.2	59.6	53.3	47.8	44.9
			MCDB	48.3	44.1	49.9	62.9	67.6	77.2	75.1	75.7	63.1	54.2	49.5	45.7
2%		WB	43.5	41.6	43.2	46.9	55.0	59.8	61.5	61.4	57.0	51.2	45.7	42.8	
		MCDB	44.0	43.0	46.4	53.6	65.4	71.9	70.7	69.5	60.3	52.5	46.8	44.6	
5%		WB	40.8	39.9	41.1	44.8	52.7	57.6	59.8	60.0	55.3	49.5	43.7	40.8	
		MCDB	41.4	41.4	43.3	49.7	61.3	65.5	66.5	66.2	58.7	51.0	44.8	42.6	
10%		WB	38.7	38.3	39.0	43.5	50.7	55.4	58.5	58.7	54.0	47.5	42.0	38.6	
		MCDB	39.2	39.5	41.1	48.1	57.4	61.0	63.6	63.5	56.8	49.0	43.2	39.5	
Mean Daily Temperature Range		5% DB	MDBR	7.6	10.9	12.3	16.2	17.7	15.0	13.1	14.1	13.5	11.4	8.2	7.7
			MCDBR	8.4	11.8	16.2	25.9	29.0	28.0	25.2	27.1	20.3	13.5	8.8	8.9
	MCWBR		6.6	8.9	11.1	15.5	15.8	14.3	13.2	14.4	12.5	9.8	6.9	7.1	
	5% WB	MCDBR	7.7	10.3	15.3	20.1	25.3	23.6	19.5	19.1	15.1	10.3	8.5	8.6	
		MCWBR	6.4	8.1	11.1	12.9	14.2	12.2	10.9	11.0	10.2	8.4	7.3	7.2	
Clear-Sky Solar Irradiance	taub	0.291	0.307	0.322	0.377	0.385	0.386	0.387	0.361	0.348	0.331	0.292	0.264		
	taud	2.166	2.290	2.329	2.209	2.305	2.373	2.413	2.514	2.519	2.418	2.238	2.133		
	Ebn at Noon	170	231	262	262	269	270	267	268	253	220	169	141		
	Edh at Noon	16	23	29	38	37	35	33	28	24	20	15	12		
All-Sky Solar Radiation	RadAvg	119	344	716	1199	1529	1534	1300	1079	687	362	160	82		
	RadStd	16	65	101	141	171	119	152	142	118	55	26	11		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	N/A	+2.11	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page